

LT Credit: Bicycle Facilities

This credit applies to

- ▶ BD+C: New Construction (1 point)
- ▶ BD+C: Core and Shell (1 point)
- ▶ BD+C: Schools (1 point)
- ▶ BD+C: Data Centers (1 point)
- ▶ BD+C: Warehouses and Distribution Centers (1 point)
- ▶ BD+C: Hospitality (1 point)
- ▶ BD+C: Retail (1 point)
- ▶ BD+C: Healthcare (1 point)

INTENT

To promote bicycling and transportation efficiency and reduce vehicle distance traveled. To improve public health by encouraging utilitarian and recreational physical activity.

REQUIREMENTS

NEW CONSTRUCTION, CORE AND SHELL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, RETAIL, HEALTHCARE

Bicycle Network

Design or locate the *project* such that a *functional entry* or bicycle storage is within a 200-yard (180-meter) *walking distance* or *bicycling distance* from a *bicycle network* that connects to at least one of the following:

- ▶ at least 10 diverse uses (see Appendix 1);
- ▶ a school or *employment center*, if the project total floor area is 50% or more residential; or
- ▶ a *bus rapid transit* stop, passenger rail station, or ferry terminal.

All destinations must be within a 3-mile (4800-meter) bicycling distance of the project boundary.

Planned bicycle trails or lanes may be counted if they are fully funded by the date of the certificate of occupancy and are scheduled for completion within three years of that date.

Bicycle Storage and Shower Rooms

Case 1. Commercial or Institutional Projects

Provide *short-term bicycle storage* for at least 2.5% of all peak visitors, but no fewer than four storage spaces per building.

Provide *long-term bicycle storage* for at least 5% of all regular building occupants, but no fewer than four storage spaces per building in addition to the short-term bicycle storage spaces.

Provide at least one on-site shower with changing facility for the first 100 regular building occupants and one additional shower for every 150 regular building occupants thereafter.

Case 2. Residential Projects

Provide *short-term bicycle storage* for at least 2.5% of all peak visitors but no fewer than four storage spaces per building.

Provide *long-term bicycle storage* for at least 15% of all regular building occupants, but no less than one storage space per three residential units.

Case 3. Mixed-Use Projects

Meet the Case 1 and Case 2 storage requirements for the nonresidential and residential portions of the project, respectively.

Large-Occupancy Projects Only:

The following guidance should be applied when determining the number of showers needed for projects with a large number of occupants:

NEW CONSTRUCTION, SCHOOLS, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, RETAIL, HEALTHCARE

Provide at least one on-site shower with changing facility for the first 100 regular building occupants and one additional shower for every 150 regular building occupants thereafter, up to 999 regular building occupants.

- ▶ one additional shower for every 500 regular building occupants, for the additional 1,000 – 4,999 regular building occupants
- ▶ one additional shower for every 1,000 regular building occupants, for the additional 5,000 + regular building occupants

CORE AND SHELL

Provide at least one on-site shower with changing facility for the first 100 regular building occupants and one additional shower for every 150 regular building occupants thereafter, up to 999 regular building occupants.

- ▶ one additional shower for every 750 regular building occupants, for the additional 1,000 – 4,999 regular building occupants
- ▶ one additional shower for every 1,500 regular building occupants, for the additional 5,000 + regular building occupants

For All Projects

Short-term bicycle storage must be within 200 feet (60meters) walking distance of any main entrance. *Long-term bicycle storage* must be within 300 feet (90 meters) walking distance of any *functional entry*. Vertical distances travelled by elevator are exempt from counting towards the walking distance requirements.

Bicycle storage capacity may not be double-counted: storage that is fully allocated to the occupants of non-project facilities cannot also serve project occupants. Indoor storage is acceptable as long as it meets the walking distance requirements. On-site bicycle sharing stations within the project boundary may count for 50% of the long-term and short-term bicycle storage space. Zero lot line projects may count publicly available bicycle parking towards their short-term storage requirements if it meets the maximum allowable walking distance.

Provide at least one on-site shower with changing facility for the first 100 regular building occupants and one additional shower for every 150 regular building occupants thereafter. Exclude patients and K-12 students from the regular building occupant count. Residential projects do not need to provide additional showers.

SCHOOLS

Bicycle Network

Design or locate the *project* such that a *functional entry* and/or bicycle storage is within a 200-yard (180-meter) *walking distance* or *bicycling distance* of a *bicycle network* that connects to either of the following:

- ▶ a *bus rapid transit* stop or passenger rail station or ferry terminal; or
- ▶ 50% of dwelling units within the school's attendance boundary.

The stops/stations or dwelling units must be within no more than a 1 1/2-mile (2400-meter) biking distance (for grades 8 and below, or ages 14 and below), and 3-mile (4800-meter) biking distance (for grades 9 and above or ages 15 and above).
Provide dedicated bicycle lanes that extend at least to the end of the school property with no barriers (e.g., fences) on school property.

Bicycle Storage and Shower Rooms

Meet storage and shower requirements for all projects and provide *long-term bicycle storage* for at least 5% of all regular building occupants (excluding students grade 3 and younger), but no fewer than four storage spaces per building.

RETAIL

Bicycle Network

Meet Bicycle Network requirements for all projects.

Bicycle Storage and Shower Rooms

Meet storage distance and shower requirements for all projects and provide at least two *short-term bicycle storage* spaces for every 5,000 square feet (465 square meters), but no fewer than two storage spaces per building.

Provide *long-term bicycle storage* for at least 5% of regular building occupants, but no fewer than two storage spaces per building in addition to the short-term bicycle storage spaces.

Provide a bicycle maintenance program for employees or bicycle route assistance for employees and customers. Route assistance must be provided in a manner easily accessible to both employees and customers.

HEALTHCARE

Bicycle Network

Meet Bicycle Network requirements for all projects.

Bicycle Storage and Shower Rooms

Meet storage and shower requirements for all projects and provide *short-term bicycle storage* for at least 2.5% of all peak visitors, but no fewer than four storage spaces per building.

Provide *long-term bicycle storage* for at least 5% of regular building occupants (excluding patients), but no fewer than four storage spaces per building in addition to the short-term bicycle storage spaces.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

This updated credit is more achievable for diverse project-types as storage and facility requirements are more representative of common site conditions. In recognition of their growing prevalence and impact, on-site bicycle sharing stations are now allowed to count for 50% of long-term and short-term bicycle storage space for all projects. Required walking distances to storage have been extended based on feedback regarding common layout conditions. To better accommodate projects with a high occupancy count, a LEED Interpretation about showers has been adopted that adjusts the number of showers that are realistic for large buildings.

Step-by-Step Guidance

Refer to the LEED v4 reference guide, with the following addition:

- ▶ Take note of the exemption of vertical distance from the walking distance requirements.
- ▶ Refer to new requirements and thresholds before proceeding with storage equations, such as the new percentage storage required per three dwelling units for residential projects.
- ▶ Step 5. For high-occupancy projects, meet the applicable shower and changing facility requirements for the building's space type.
- ▶ For multifamily and residential projects, the ability to store bicycles within units does not count as long-term storage.

Further Explanation

Refer to the LEED v4 reference guide, with the following additions:

Selecting Bicycle Storage

Recommended bicycle rack designs, derived from The Association of Pedestrian and Bicycle Professionals (APBP) Essentials of Bike Parking Guide, include:

- ▶ For all applications: Inverted U (or Stable, Loop), Post and Ring, and Wheelwell-secure.
- ▶ For high- density, space-constrained situations: Staggered Wheelwell-secure, Vertical, and Two-Tier.
- ▶ Bicycle rack designs that are not recommended include: Wave, Schoolyard, Coathanger, Wheelwell, Bollard, Spiral, and Swing Arm Secured.

Rating System Variations

Refer to the LEED v4 reference guide, with the following addition and modifications:

Core and Shell

Refer to Appendix 2, Default Occupancy Counts, for occupancy count requirements and guidance.

Retail

For projects that are part of a multitenant complex only: If bicycle storage spaces have been provided in the complex in which the project is located, determine the number of spaces that may be attributed to the project by dividing the project's floor area by the total floor area of the development (buildings only) and multiplying the percentage result by the total number of spaces. If this number does not meet the credit requirement, the project must provide additional bicycle storage.

Historic Urban Locations

If the requirements on the width of bike paths cannot be met due to the *historic urban context* of the bicycle network, compensating measures to reduce street speeds and/or to enhance biking security on routes connecting to a qualifying bike network are acceptable:

- ▶ A security lane for biking (marked dedicated bike lane, which can be shared by cars in narrow sections of the street when no bikes are present) or a physically dedicated bike lane less than 5 feet (1.5 meters) wide
- ▶ Intersections spaced no more than 400 feet (122 meters) apart
- ▶ Travel lane widths no greater than 10 feet (3 meters) and parallel parking lane widths no greater than 8 feet (2.4 meters)

Historicity of the area and/or whether the site receives legal protection are determining factors for whether a building has a historic urban context.

Referenced Standards

The Association of Pedestrian and Bicycle Professionals (APBP), Essentials of Bike Parking: Selecting and Installing Bike Parking that Works (2015), pages 6-8:

https://www.apbp.org/resource/resmgr/Bicycle_Parking/EssentialsofBikeParking_FINAL.pdf

Required Documentation

Refer to the LEED v4 reference guide, with the following addition:

Historic urban locations and routes must be clearly identified by type in a narrative.

Definitions

- ▶ **long-term bicycle storage** bicycle parking that is easily accessible to residents and occupants and covered to protect bicycles from rain and snow.
- ▶ **historic urban context** refers to limiting conditions linked to historic urban planning that may consequentially impact buildings and infrastructure within the associated jurisdiction. Such site conditions may make buildings act as traffic calming structures or may affect street access and the width of right-of-way. Historicity of the area and whether the site receives legal protection are determining factors for whether a bicycle network has a historic urban context.

Connection to Ongoing Performance:

- ▶ LEED O+M LT prerequisite Transportation Performance: Improving bicycle facilities and access to a bicycle network as well as implementing any corresponding bicycle programs may help improve a project's transportation performance score.